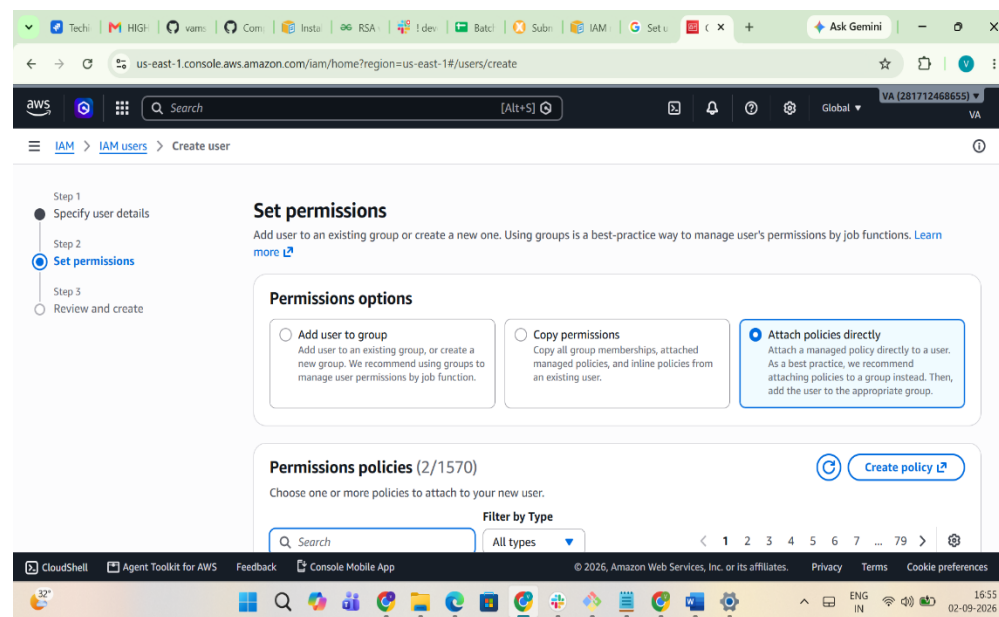


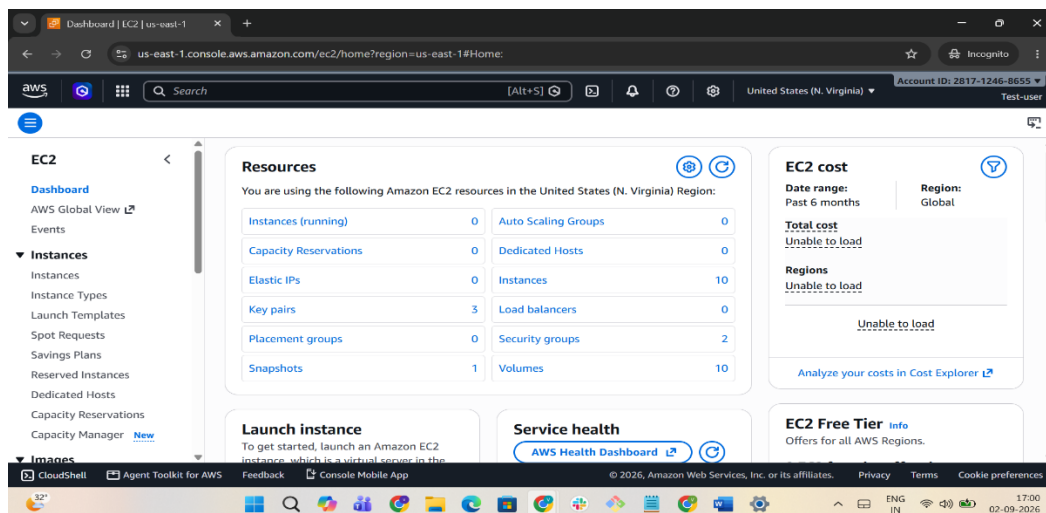
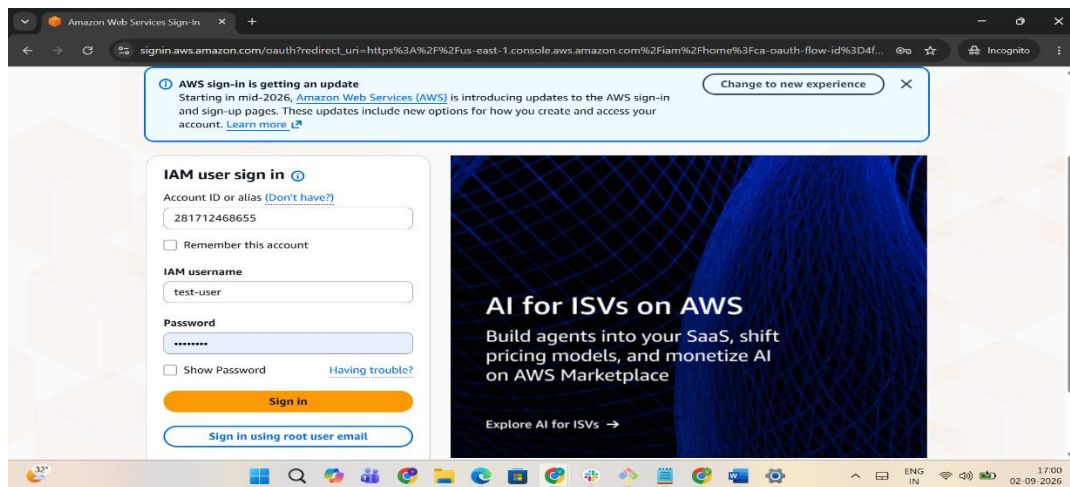
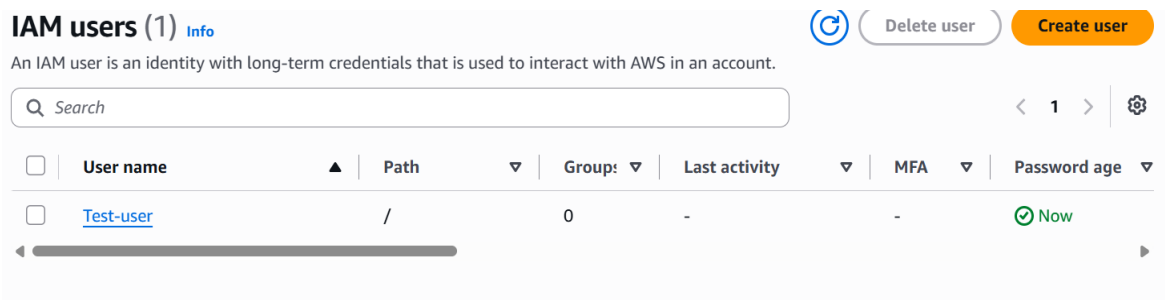
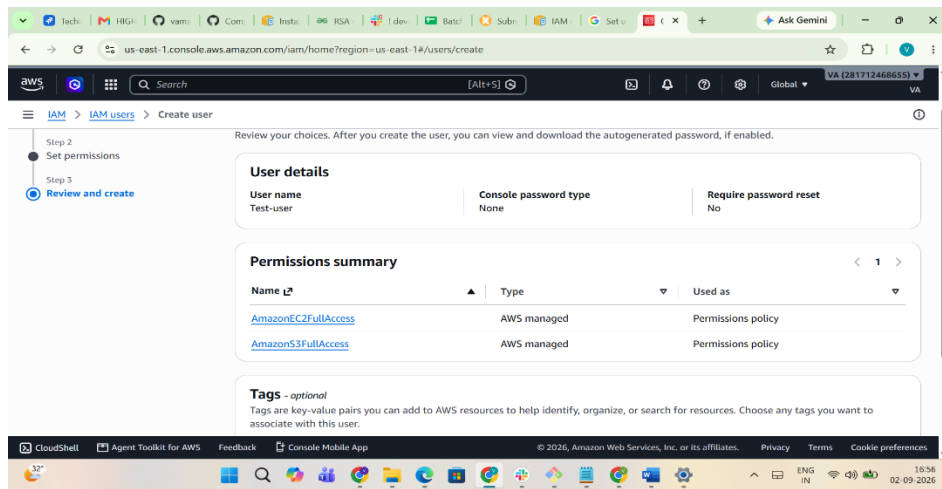
Task on IAM

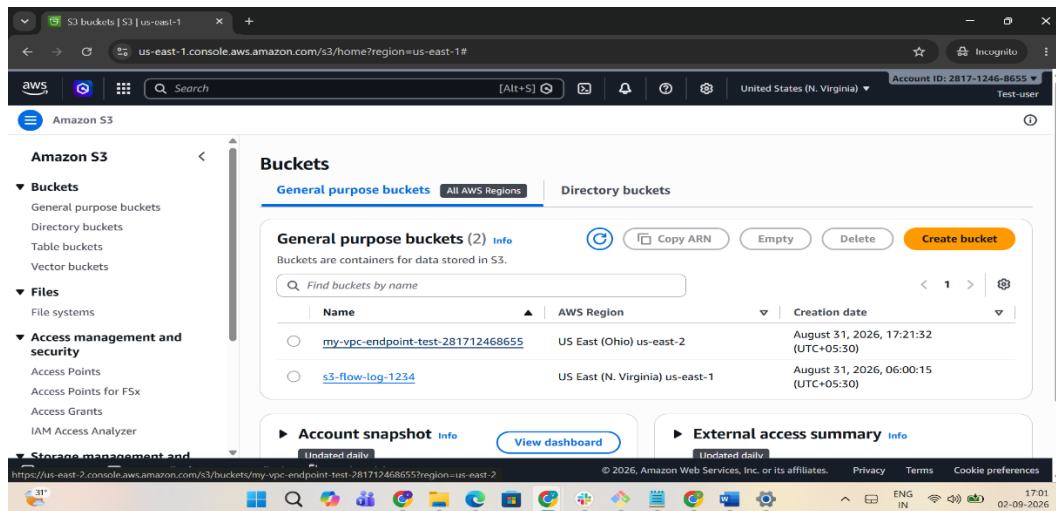
1. Create one IAM user and assign EC2 and S3 full access roles.

Implementation

1. Open the AWS Management Console and navigate to IAM.
2. Go to Users → Create user.
3. Enter a username, for example, Test-user.
4. Enable AWS Management Console access if required.
5. Create the user.
6. Open the created user and go to the Permissions tab.
7. Click Add permissions → Add permissions directly.
8. Attach the AmazonEC2FullAccess policy.
9. Attach the AmazonS3FullAccess policy.
10. Review and save the permissions.
11. Verify that the user has full access to EC2 and S3 services.



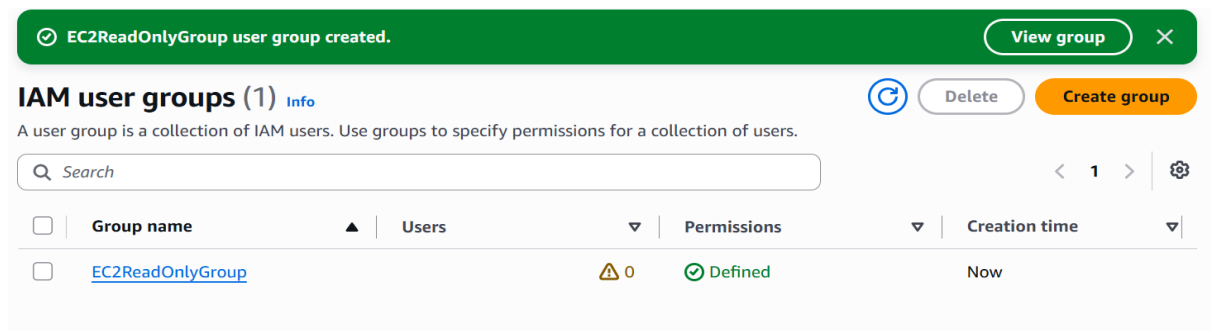


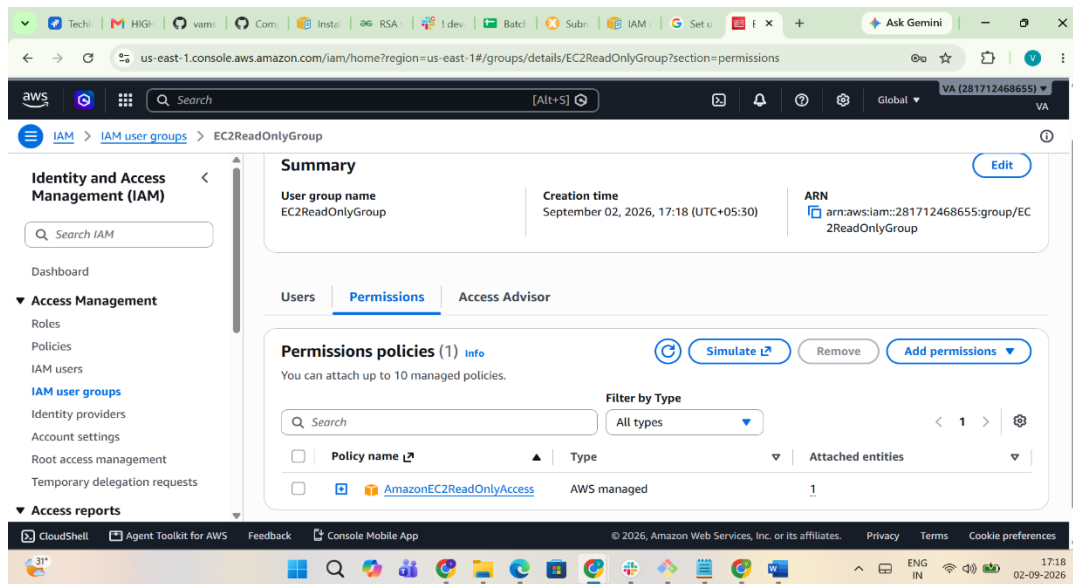


2. Create one group in IAM and assign read access for EC2.

Implementation

1. Open IAM → User groups.
2. Click Create group.
3. Enter the group name as EC2ReadOnlyGroup.
4. Search for the AmazonEC2ReadOnlyAccess policy.
5. Select the policy.
6. Create the group.
7. Verify that the AmazonEC2ReadOnlyAccess policy is attached to the group.
8. Users added to this group will automatically receive EC2 read-only permissions.

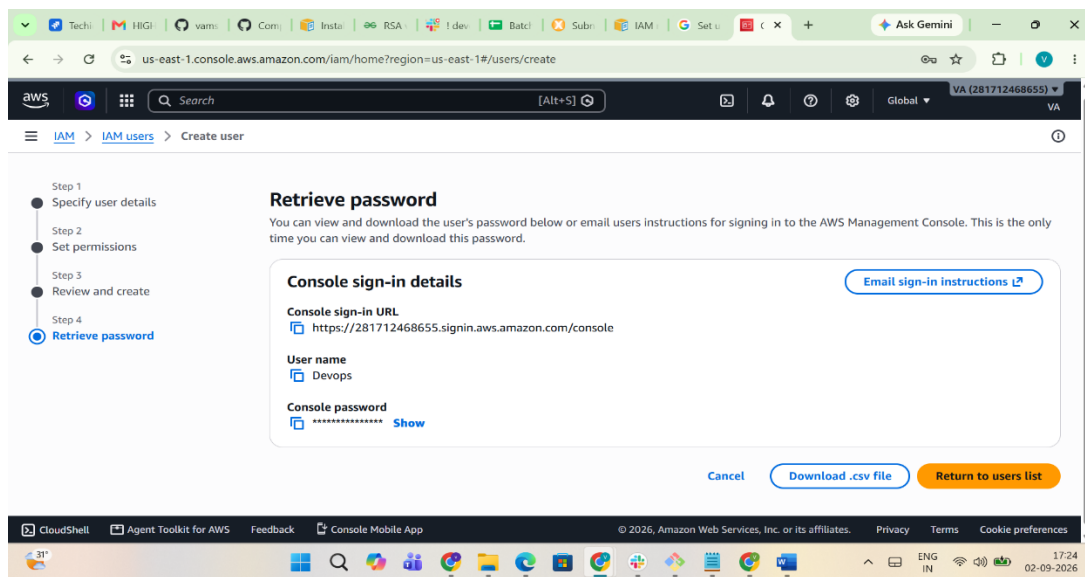
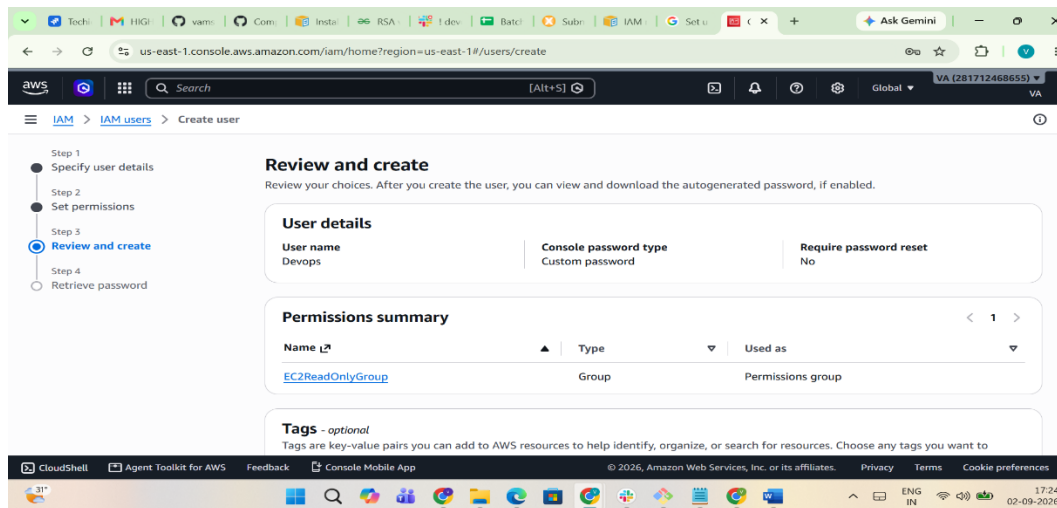
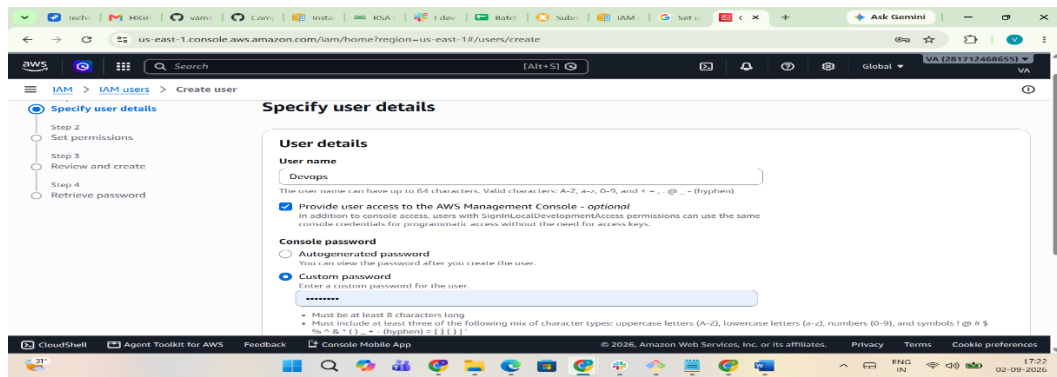


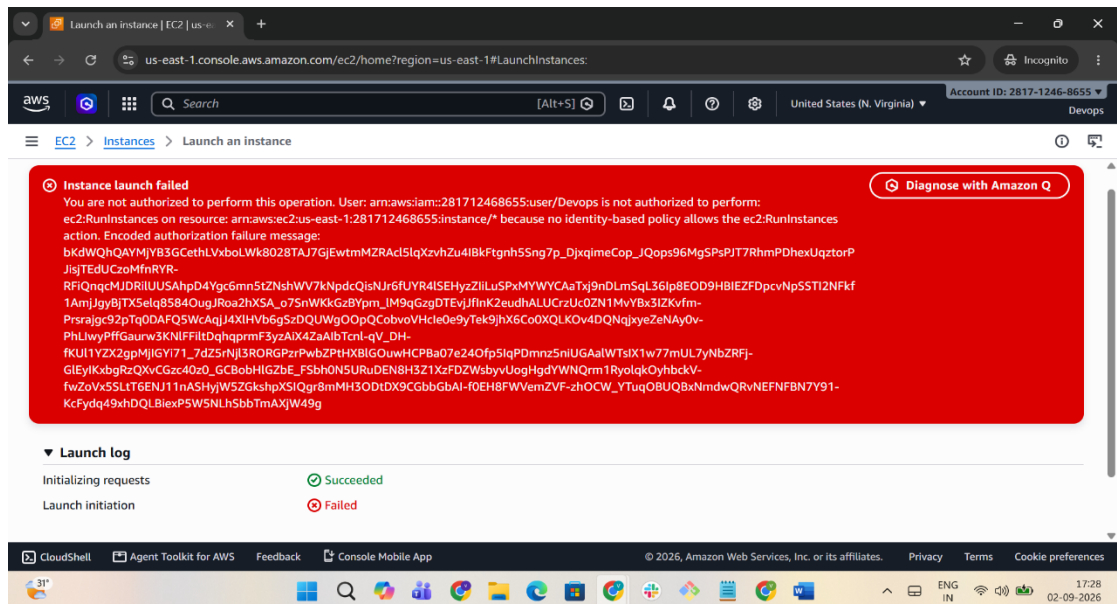
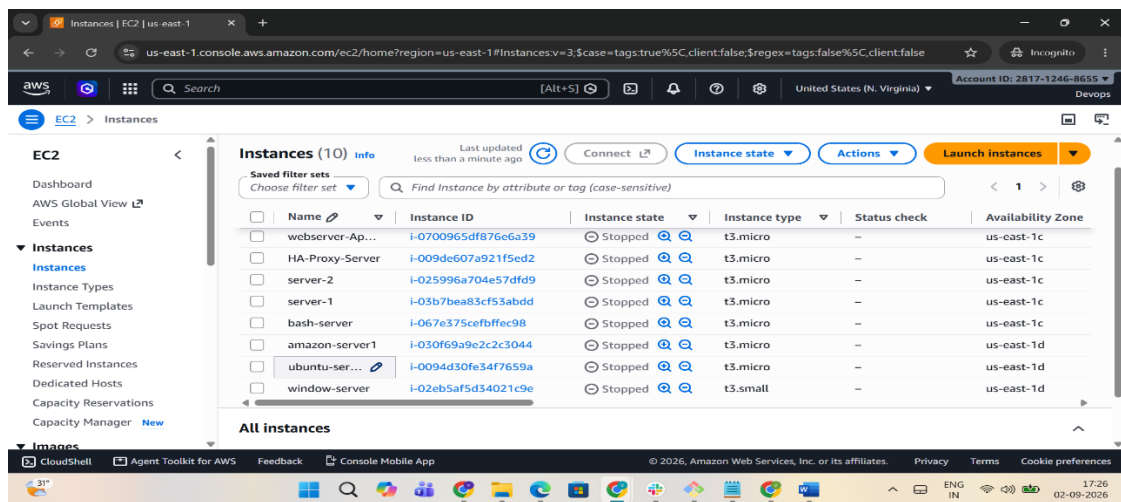
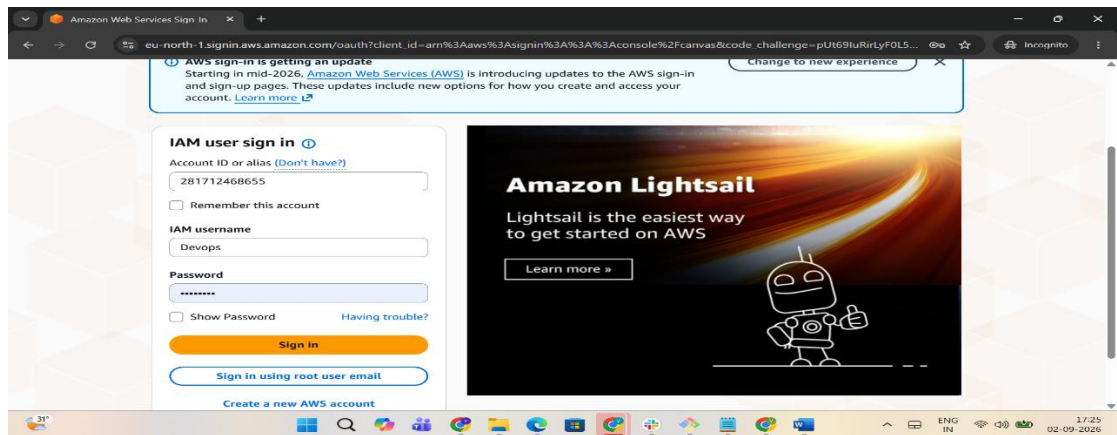


3. Create a new user named "Devops" and add to the group created in task 2.

Implementation

1. Open IAM → Users → Create user.
2. Enter the username as Devops.
3. Enable console access if required.
4. Create the user.
5. Open the Devops user.
6. Select Add permissions → Add user to group.
7. Select the EC2ReadOnlyGroup.
8. Add the user to the group.
9. Verify that Devops is listed as a member of the group.
10. Test the user and verify that EC2 resources can be viewed but cannot be modified.





4. Write a bash script to create an IAM user with VPC full access.

Implementation

1. Install and configure the AWS CLI.

- ```
#!/bin/bash
```

```
aws iam create-user --user-name "$USER_NAME"
```

```
aws iam attach-user-policy \
```

```
--user-name "$USER_NAME" \
```

```
--policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess
```

```
echo "IAM user $USER_NAME created with VPC Full Access."
```

7. Give execute permission to the script:

```
chmod +x create-vpc-user.sh
```

8. Create a role with IAMFullAccess policy AND then add to ec2-instance IAMrole

9. Run the script:

```
./create-vpc-user.sh
```

9. Verify the attached policy:

```
aws iam list-attached-user-policies --user-name VPCAdmin
```

10. Confirm that AmazonVPCFullAccess is attached to the newly created IAM user.

```
91738@IDEAPAD MINGW64 ~/Downloads
$ ssh -i ohio-key.pem ec2-user@18.117.160.213
The authenticity of host '18.117.160.213 (18.117.160.213)' can't be established.
ED25519 key fingerprint is SHA256:oRXBqmtyg+4ZFqhWmE1/O2tYNNj2gJQsf1vDkQOWZOU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.117.160.213' (ED25519) to the list of known hosts.
```



```
Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-20-114 ~]$ aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.18.44-99.149.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-172-31-20-114 ~]$ aws configure
```

```
[root@ip-172-31-20-114 ~]# vi create-vpc-user.sh
[root@ip-172-31-20-114 ~]# cat create-vpc-user.sh
#!/bin/bash

USER_NAME="VPCAdmin"

aws iam create-user --user-name "$USER_NAME"

aws iam attach-user-policy \
 --user-name "$USER_NAME" \
 --policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess

echo "IAM user $USER_NAME created with VPC Full Access."
[root@ip-172-31-20-114 ~]# chmod +x create-vpc-user.sh
```

```
root@ip-172-31-20-114:~
#!/bin/bash

USER_NAME="VPCAdmin"

aws iam create-user --user-name "$USER_NAME"

aws iam attach-user-policy \
 --user-name "$USER_NAME" \
 --policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess

echo "IAM user $USER_NAME created with VPC Full Access."
~
~
~
~
~
```

```
[root@ip-172-31-20-114 ~]# aws sts get-caller-identity
{
 "UserId": "AROAUDF27EKX5KDR7LLXM:i-0a2ed47555fc327ff",
 "Account": "281712468655",
 "Arn": "arn:aws:sts::281712468655:assumed-role/EC2-IAM-Admin-Role/i-0a2ed47555fc327ff"
}
[root@ip-172-31-20-114 ~]# ./create-vpc-user.sh
{
 "User": {
 "Path": "/",
 "UserName": "VPCAdmin",
 "UserId": "AIDAUDF27EKXUICVRSIAL",
 "Arn": "arn:aws:iam::281712468655:user/VPCAdmin",
 "CreateDate": "2026-09-02T12:31:20+00:00"
 }
}
IAM user VPCAdmin created with VPC Full Access.
[root@ip-172-31-20-114 ~]# aws iam list-attached-user-policies --user-name VPCAdmin
{
 "AttachedPolicies": [
 {
 "PolicyName": "AmazonVPCFullAccess",
 "PolicyArn": "arn:aws:iam::aws:policy/AmazonVPCFullAccess"
 }
]
}
```

The screenshot displays the AWS IAM console interface. The top navigation bar shows the user is logged in as 'VA (281712468655)'. The main content area is titled 'IAM > IAM users > VPCAdmin'. On the left, there is a sidebar with 'Identity and Access Management (IAM)' and various navigation links like 'Dashboard', 'Access Management', and 'Access reports'. The main panel shows the user's details: 'arn:aws:iam::281712468655:user/VPCAdmin', 'Disabled' status, and creation date 'September 02, 2026, 18:01 (UTC+05:30)'. The 'Permissions' tab is active, displaying 'Permissions policies (1)'. A table lists the attached policy: 'AmazonVPCFullAccess', which is 'AWS managed' and 'Attached via Directly'.

## **5. Create an IAM policy to allow EC2 access for a specific user in specific regions only.**

### **Implementation**

1. Open IAM → Policies.
2. Click Create policy.
3. Select the JSON policy editor.
4. Define the required EC2 permissions.
5. Use the ec2:Region condition to restrict access to specific AWS Regions.
6. For example, allow EC2 access only in Mumbai (ap-south-1) and Singapore (ap-southeast-1).
7. Create the policy.
8. Attach the policy to the required IAM user.
9. Test EC2 access in the allowed regions.
10. Try accessing EC2 from another region and verify that access is denied.

Example policy:

```
{
 "Version": "2012-10-17",
 "Statement": [
 {
 "Effect": "Allow",
 "Action": "ec2:*",
 "Resource": "*",
 "Condition": {
 "StringEquals": {
 "ec2:Region": [
 "ap-south-1",
 "ap-southeast-1"
]
 }
 }
 }
]
}
```

```
}

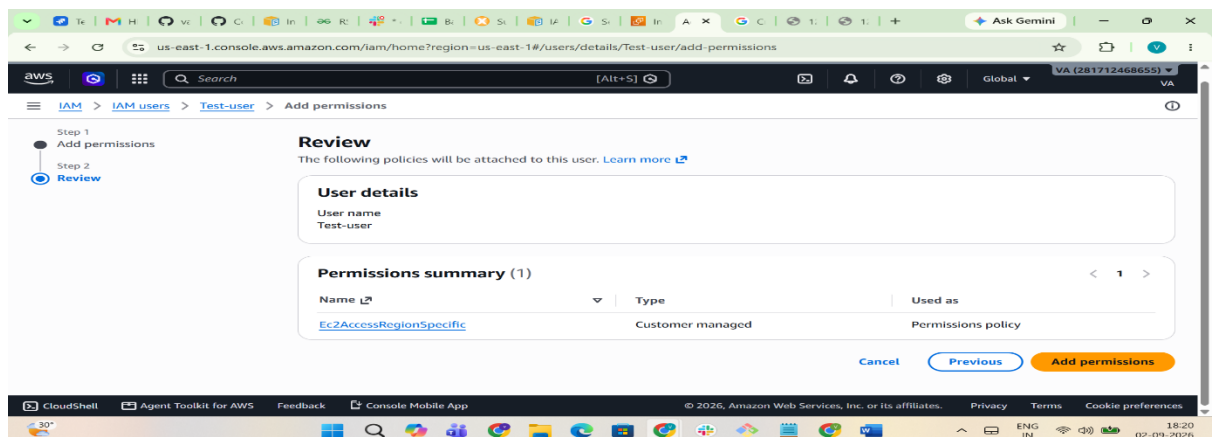
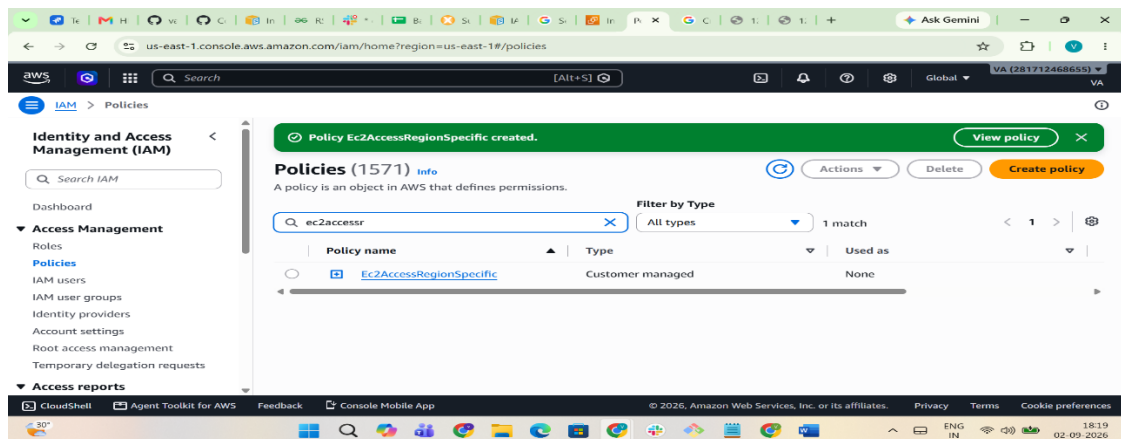
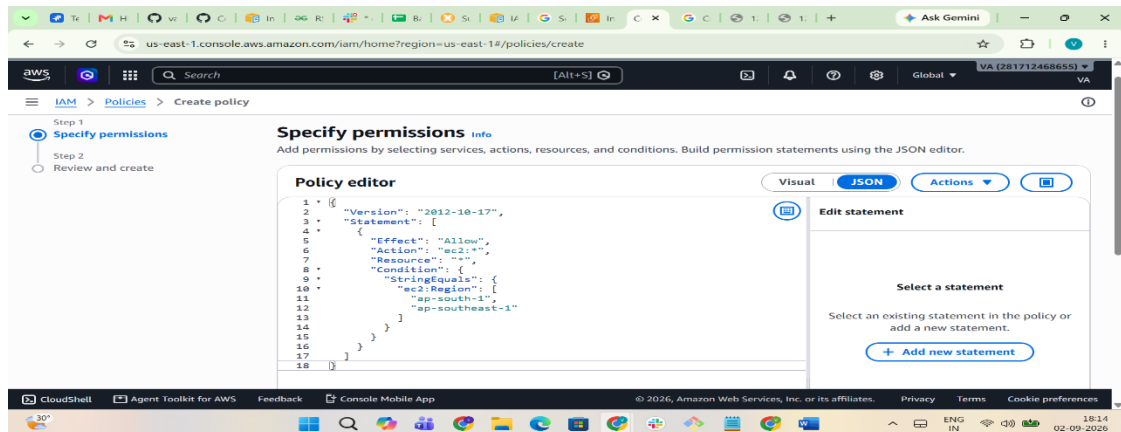
}

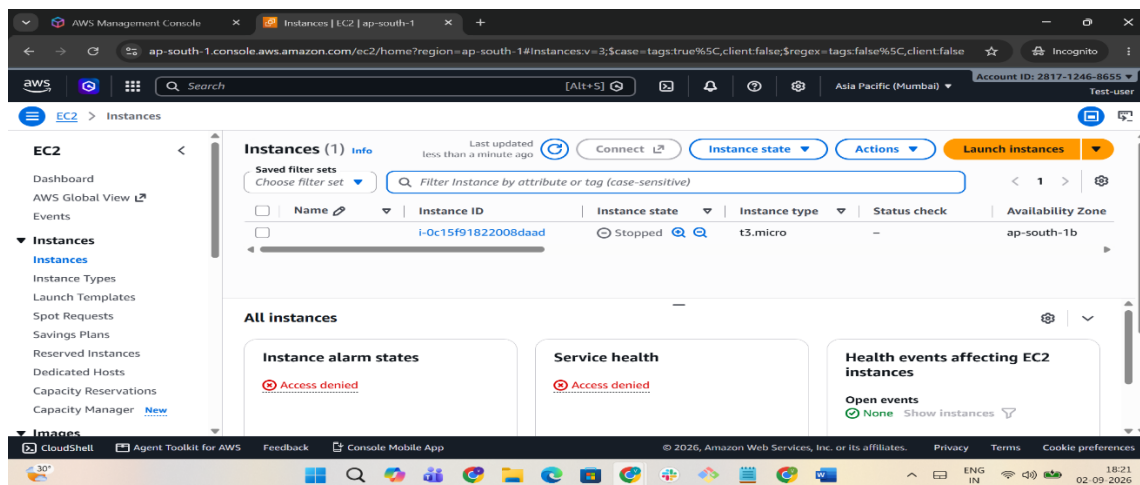
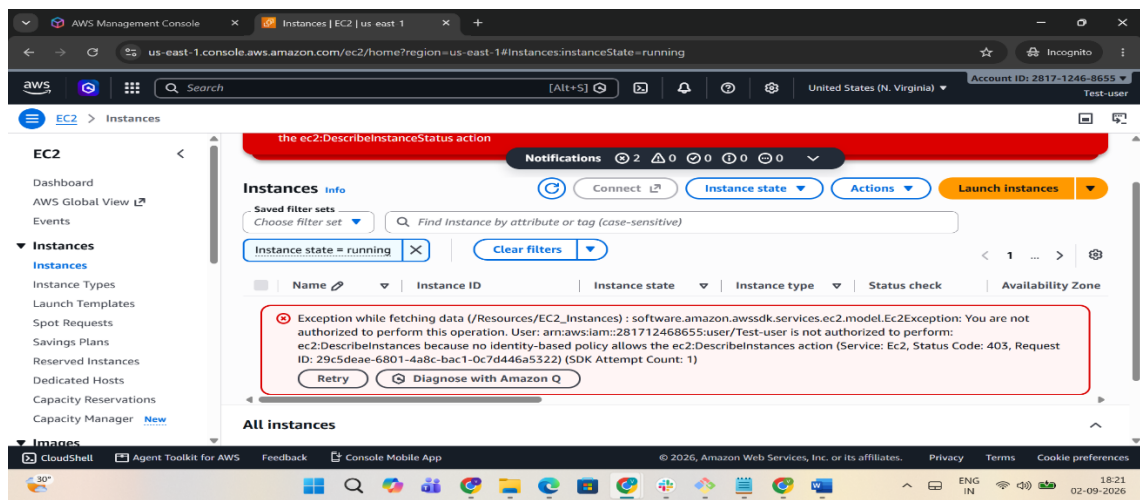
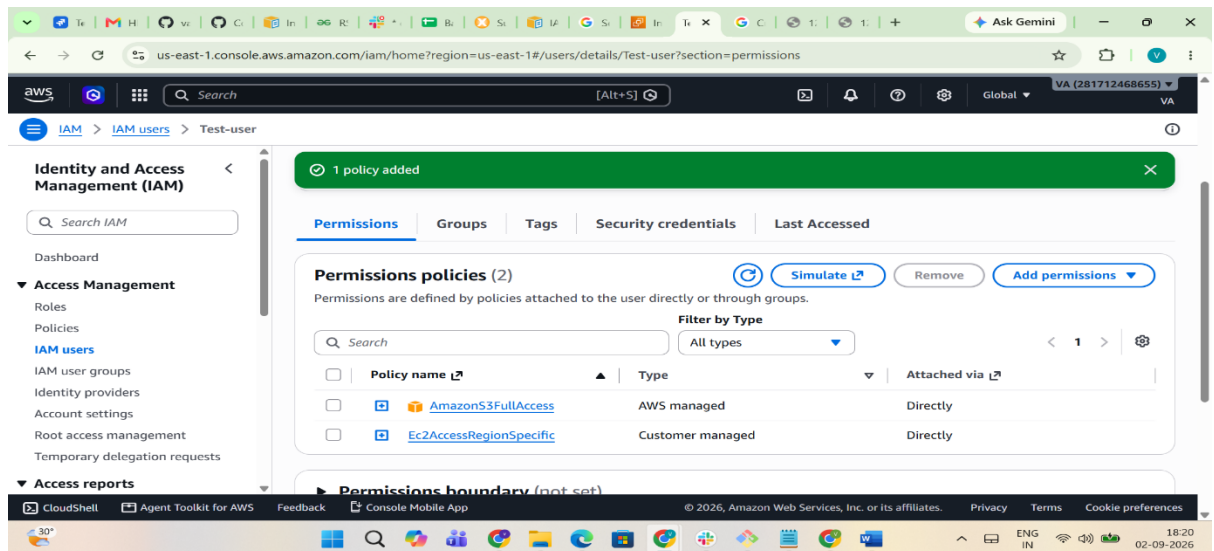
}

]

}
```

11. Verify that the user can perform EC2 operations only in the specified regions.





6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.

Implementation

Account A: Contains the IAM user Test-user.

Account B: Contains the S3 bucket.

Account B – Configure S3 Bucket

1. Log in to Account B.
2. Open S3 and create or select an existing bucket.
3. Note the bucket name, for example account-b-devops-bucket.
4. Open the bucket and go to Permissions.
5. Open Bucket Policy.
6. Specify the IAM user from Account A as the principal.
7. Grant the required permissions such as s3:ListBucket, s3:GetObject, and s3:PutObject.
8. Save the bucket policy.

Example:

```
{
 "Version": "2012-10-17",
 "Statement": [
 {
 "Sid": "AllowAccountAUser",
 "Effect": "Allow",
 "Principal": {
 "AWS": "arn:aws:iam::ACC_ID:user/Test-user"
 },
 "Action": [
 "s3:ListBucket"
],
 "Resource": "arn:aws:s3:::account-b-devops-bucket-050910"
 },
 {
 "Sid": "AllowAccountAUserObjects",
 "Effect": "Allow",
```

```
"Principal": {
 "AWS": "arn:aws:iam::acc_ID:user/Test-user"
},
"Action": [
 "s3:GetObject",
 "s3:PutObject"
],
"Resource": "arn:aws:s3:::account-b-devops-bucket-050910/*"
}
]
```

}Account A – Configure IAM User

9. Log in to Account A.
10. Open IAM → Users → Test-user.
11. Attach an IAM policy that allows the required S3 actions.
12. Specify the Account B S3 bucket as the resource.
13. Save the policy.
14. Configure AWS CLI using the Account A Devops credentials.
15. Test access to the Account B bucket:  
`aws s3 ls s3:// account-b-devops-bucket-050910`
16. Upload a test file:  
`aws s3 cp test.txt s3:// account-b-devops-bucket-050910/`
17. Verify the uploaded file from Account B.
18. Confirm that the Account A user can access the S3 bucket owned by Account B.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/users/details/Test-user?section=permissions

aws IAM IAM users Test-user

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

ARN: [arn:aws:iam::281712468655:user/Test-user](#)

Created: September 02, 2026, 16:57 (UTC+05:30)

Console access: [Enabled without MFA](#)

Last console sign-in: [Today](#)

Access key 1: [Create access key](#)

Permissions Groups Tags Security credentials Last Accessed

Permissions policies (2)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

| Policy name                        | Type        | Attached via |
|------------------------------------|-------------|--------------|
| <a href="#">AmazonS3FullAccess</a> | AWS managed | Directly     |

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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28° ENG IN 17:23 03-09-2026

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-02e89d5c1ccc25108

aws EC2 Instances i-02e89d5c1ccc25108

EC2

Dashboard

AWS Global View

Events

Instances

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- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager

Images

- AMIs
- AMI Catalog

Instance summary for i-02e89d5c1ccc25108 (ec2-ohio)

Updated less than a minute ago

Instance ID: [i-02e89d5c1ccc25108](#)

IPv6 address: -

Hostname type: IP name: ip-172-31-39-67.us-east-2.compute.internal

Answer private resource DNS name: IPv4 (A)

Auto-assigned IP address: [18.226.230.208](#) [Public IP]

Public IPv4 address: [18.226.230.208](#) | [open address](#)

Instance state: [Running](#)

Private IP DNS name (IPv4 only): [ip-172-31-39-67.us-east-2.compute.interna](#)

Instance type: [t3.micro](#)

VPC ID: [vpc-0926f9e7dd991735e](#) (Default-vpc)

Private IPv4 addresses: [172.31.39.67](#)

Public DNS: [ec2-18-226-230-208.us-east-2.compute.amazonaws.com](#) | [open address](#)

Elastic IP addresses: -

AWS Compute Optimizer finding: [Opt-in to AWS Compute Optimizer for recom](#)

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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28° ENG IN 18:09 03-09-2026

```
root@ip-172-31-39-67:~
$ cat test.txt
hello, welcome to ubuntu

root@ip-172-31-39-67:~/Downloads
$ vi test.txt

root@ip-172-31-39-67:~/Downloads
$ ssh -i ohio-key.pem ec2-user@18.226.230.208
The authenticity of host '18.226.230.208 (18.226.230.208)' can't be established.
ED25519 key fingerprint is SHA256:31j8rlecFNLyapXfYWKIwlhuvuQ81vKb1GgtdsfmQA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.226.230.208' (ED25519) to the list of known hosts

--\----- Amazon Linux 2023
--\-----
--\-----
--\----- https://aws.amazon.com/linux/amazon-linux-2023
--\-----
--\-----
--\-----
[ec2-user@ip-172-31-39-67 ~]$ sudo su -
[root@ip-172-31-39-67 ~]# aws sts get-caller-identity

Unable to locate credentials. You can configure credentials by running "aws login".
[root@ip-172-31-39-67 ~]# aws configure
AWS Access Key ID [None]: AKIAUDF27EKXYVTLBND
AWS Secret Access Key [None]: mlal1UGV9VTSy1/Y/MleQEAmGPDDwdGQUkP0wOvc
Default region name [None]: us-east-2
Default output format [None]: json
[root@ip-172-31-39-67 ~]# aws sts get-caller-identity
{
 "UserId": "AIDAUDF27EKXWzGBRBTZ",
 "Account": "281712468655",
 "Arn": "arn:aws:iam::281712468655:user/Test-user"
}
[root@ip-172-31-39-67 ~]# aws s3 ls
2026-09-03 12:12:44 account-b-devops-bucket-0509
2026-09-03 11:05:37 my-devops-demo0509
2026-08-31 11:51:24 my-vpc-endpoint-test-281712468655
2026-08-31 00:35:04 s3-flow-log-1234
[root@ip-172-31-39-67 ~]#
[root@ip-172-31-39-67 ~]# aws s3 ls s3://account-b-devops-bucket-050910/
2026-09-03 12:10:57 Demo file.txt
[root@ip-172-31-39-67 ~]#
```